Let’s do the ScoreWarp again! Shifting notes to performance timelines

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Introduction

As per Hargreaves et al (2005), music is a fundamental channel of communication in which expressive intentions and meanings flow from a composer’s intention to a performer’s intention, usually using the score as a medium, and from there via the performer’s interpretive actions to a listener’s perception. The existence of critical score editions implies a secondary target audience for music notation: not only performers, but also music scholars, scientists, and others interested in a critical analytical understanding of the music.

Music is typeset accordingly to cater to these two use-cases: performance scores are designed with the musician in mind, with care taken in adjusting the placement of system- and page-breaks, the density of notation on page, and numerous other layout-specific factors to optimise performance comfort. Meanwhile, critical score editions are designed to best convey music-semantic factors of interest, varying significantly based not only on the material conveyed but on the specific target focus – with genetic edition, e.g., primarily concerned with the process of composition, even more than with the musical content itself. For these reasons, critical editions are often unsuitable, if not entirely incompatible, with music performance.

A third use-case of music notation involves the injection of music semantics into extra-musical information contexts. This applies particularly where the phenomenon of

primary interest unfolds over a (physical) timeline, but reflects or is influenced by a musical structure. Examples abound in performance science, music psychology, and other empirical research on performing arts that capture human behavioural or physiological parameters over time in the presence of a performed musical score.

1 Visualising timed information in alignment with scores

Prior research has investigated the alignment of music scores and performance timelines from different perspectives: alignment algorithms have become a core topic of research in music informatics (Dixon & Widmer, 2005; Nakamura et al, 2017; Müller et al, 2021); whereas the modelling of such aligned timelines presents an interesting problem for digital musicology (Fields et al, 2011; Nurmikko-Fuller et al, 2015). The issue of how to visualise such alignments has received less formal attention, though approaches may be roughly categorised according to the presence and use of a (physical- or score-) time axis, as follows:

No time axis. In this approach often taken when visualising performance characteristics or listener behaviours (e.g., Langner & Goebl, 2003), no time axis is visualised; instead, observations are plotted against axes representing the dependent variables of interest, with time represented through animation and/or through the use of colouring to indicate temporal currency (e.g., by gradually fading out older observations as they are replaced by new ones). Alternatively, data may be plotted against an abstracted structural representation of the score, preserving ordering but otherwise omitting temporal information (e.g., mean valence/arousal responses to specified music segments as in Yang et al, 2021).

Temporal information displayed against a score-time axis. Here, measurements taken in physical time are displayed in positional alignment with a musical score, such that observations are plotted in alignment with the score positions they coincide with – either through a visually displayed score, or simply against an axis running proportionally in score-time from the beginning to the end of the piece or excerpt of interest. This visually warps the physically measured timeline to coincide with a depicted score time. Examples include tempo or inter-onset interval curves such as in Gabrielsson (1983) or Repp (1998).

Physical-time axis with visual score-time alignment indicators. Typically this is achieved by using angled lines to connect time instants of interest from one or several visualisations of physically-timed information to corresponding elements in a displayed score (Flossmann et al, 2010; Goebl & Palmer, 2013). Alternatively, score fragments may simply be visually juxtaposed in approximate alignment with the physical time, either as an additional horizontal layer (e.g., Goebl, Flossmann, & Widmer, 2010) or as small fragments associated with highlighted time intervals (e.g., Egermann et al, 2013).

Implicit physical timeline through playback (“*score following*”). Alternatively, an implicit physical timeline may unfold alongside a depicted score, through the playback of a corresponding MIDI rendition or performance audio recording, with the alignment typically visualised using automated paging or scrolling of the score, via an animated cursor, and/or through visual highlighting of score elements, such as in Freischütz digital

(Röwenstrunk et al, 2015) or the playback functionality of the MEI Viewer[1], mei-friend[2] (Goebl & Weigl, 2024), or the Verovio Humdrum Viewer[3].

Combined approaches. Different aspects of the above strategies may be combined in order to provide synoptic views connecting physical- or score-timelines with score depictions, typically including score following. For example, the TuttiTempi performance comparison tool (Viro, 2021) warps timed information to fit a score-time axis in juxtaposition with an automatically scrolling and highlighting score. In the performance comparison tool "Listen Here!" (Weigl et al, 2023), multiple physical performance timelines are aligned through the placement of markers or a playback cursor (akin to 'score following', but in absence of a visible score); additionally, score annotations may be imported into the application, and are projected into physical time through highlighting of the corresponding intervals on each performance timeline.

2 Warping the score

Here, we propose a new approach, thus far largely neglected, we suspect, due to the previously-high level of technical difficulty in its implementation: to display timed information on a physical time axis, but to depict this using a warped version of the music score, in which elements are physically shifted so that their location linearly coincides with the physical time of their sounding.

When performing such a warping operation globally over all elements within a score system, ordering along the horizontal axis is retained. The warped score thus remains a reliable source of music semantics, but conveys accurate temporal information in physical time. It is of course no longer optimised for performance, nor necessarily for music-critical scholarship; instead, it becomes available as a music-semantic visual index into timed data.

Such an approach has been previously approximated by juxtaposing piano-roll-like notations with score fragments (Nettheim, 2001). The proposed warping process obviates the need for the additional piano-roll element and thus simplifies the cognitive task of understanding the position of performed notes in physical time. We have implemented this approach on MEI-encoded scores rendered using Verovio (Pugin et al, 2014), with the toolkit's semantically structured SVG output greatly simplifying the task of performing this operation.

Our ScoreWarp library takes two files as input: a single-system score SVG object rendered from an MEI file with the help of the Verovio engraving library, and a MAPS alignment file (Weigl, 2020). The latter contains the results of a matching process with an entry for each note element in the score pointing to the corresponding performance timestamp. Multiple notes with an identical timestamp are represented by the same entry. For our demo, the alignment is achieved using the MIDI-to-MIDI matching algorithm by Nakamura et al (2017) with post-processing to match score-time points to MEI note identifiers. Figure 1a shows a score excerpt with lines connecting the note positions with the performance timeline.

ScoreWarp takes the horizontal coordinates of the first and the last note in the SVG score graphic and uses these to scale the first and last performance timestamp within the MAPS

[1] https://editor.verovio.org/ (accessed 22 April 2025)
[2] https://mei-friend.mdw.ac.at/ (accessed 22 April 2025)
[3] https://verovio.humdrum.org/ (accessed 22 April 2025)

alignment data. It creates a mathematical function to linearly interpolate the intermittent timestamps in the MAPS file in order to proportionally shift elements that do not strictly coincide temporally with a timestamp, such as barlines, hairpins, directives, etc. Having executed the `warp()` function (see Figure 1b), it proceeds through a list of SVG elements, horizontally shifting (and where required, scaling) each element according to the mathematical function, by adding a SVG `@transform` attribute. The first note inside a chord is used to shift the entire chord element. Spanning elements such as slurs or beams are shifted and scaled relative to the corresponding notehead coordinates. The ledger lines are shifted, making use of the `@data-related` attribute through the `svgHtml5` option of the Verovio toolkit to associate them with their corresponding noteheads.

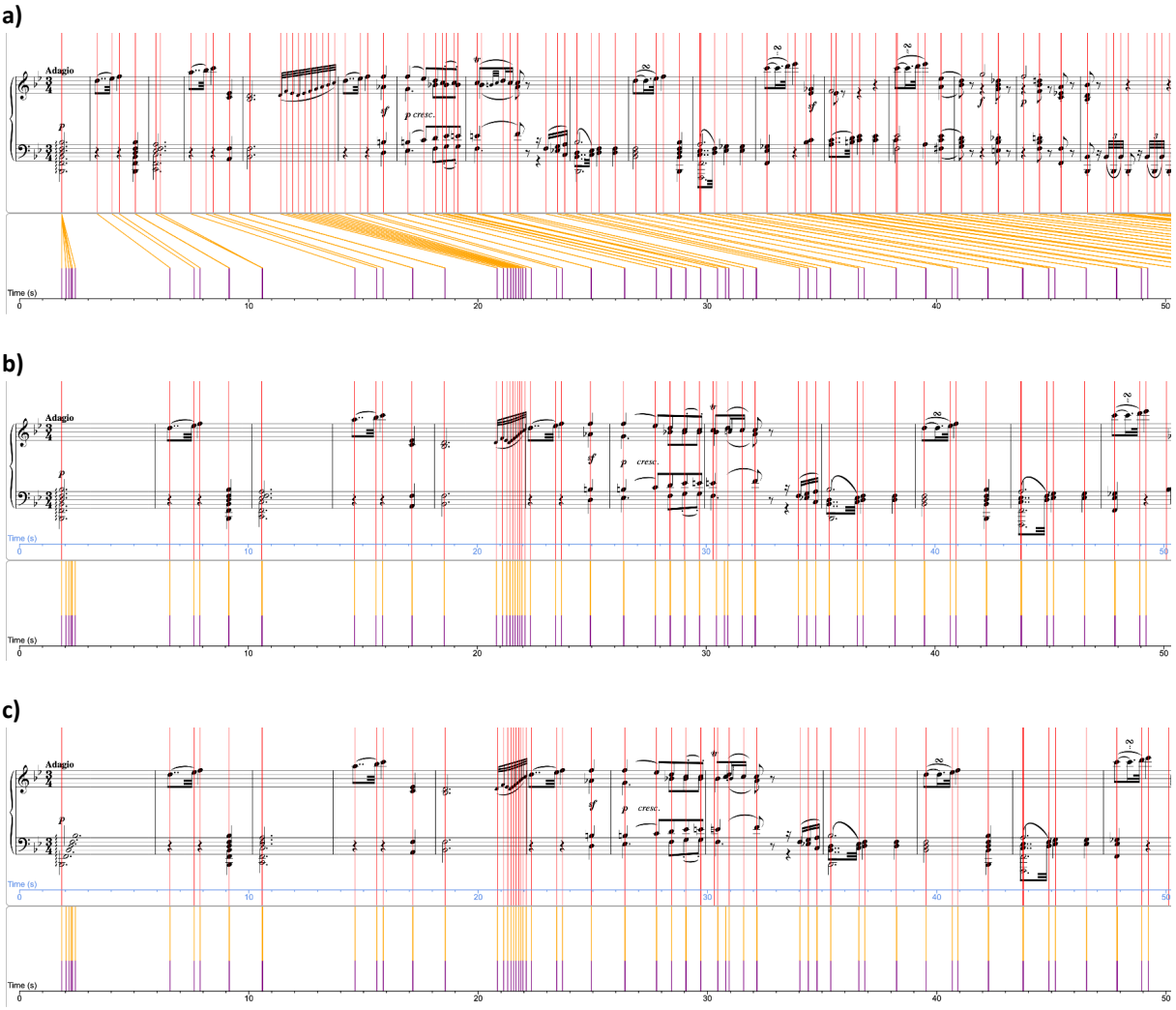

Figure 1: Screenshots of the beginning of the second movement of Beethoven's Sonata Op. 31 No. 2 showing a score timeline (marked by red vertical lines) and performance timeline (purple lines) connected by orange lines. **a)** Unaltered score with orange lines connecting score timeline elements with corresponding performance timeline instants. **b)** Warped score matching the performance timeline; chords are shifted to match the instant of their first sounded note. **c)** Score warped on an individual note level, visually reflecting arpeggios or other performed asynchronies. Please note here the spread of the score notes in the initial arpeggiated chord in the second staff.

While the `warp()` function shifts all notes of a chord to a single timestamp, performance data available from recordings on MIDI-enabled instruments (e.g., Yamaha Disklavier, Bösendorfer CEUS) contain fine-grained performance timing data for each note. To make full use of this additional detail, the `warpIndividualNotes()` function adjusts the position of each note to the corresponding timestamp, visualising performance effects like arpeggios or the melody lead phenomenon (Repp, 1996, see Figure 1c). ScoreWarp finally adds a physical-time axis to the warped SVG graphic as a separate group element (`<g>`), providing a readily-available alignment target for time-series data which may be included in post-processing.

ScoreWarp is available on a demonstration page at https://iwk-digital.github.io/scorewarp/ featuring several example pieces and performances. The demo page exposes the functions of ScoreWarp to the user and provides download access to the warped SVG file for inclusion in other applications. It also allows users to upload their own files, consisting of one MEI file and one or multiple MAPS files, each corresponding to a performance. These are then warped in batch, with the resulting SVGs provided for download as a zip file. The codebase is available under the GNU AGPL v3.0 license at https://github.com/iwk-digital/scorewarp.

3 Conclusion and future work

We have proposed a novel approach to solving the problem of visualizing information captured according to physical- and score-time modalities. Rather than relying on visual or auditory-visual juxtaposition, or warping physical-timeline information to suit a particular score layout, we instead warp the positioning of notation elements within a score to fall onto coinciding physical-temporal positions. We have presented a feasibility evaluation of the proposed approach in the open-source ScoreWarp prototype, which should allow relatively simple reuse of the idea within external applications. We plan to extend the set of compatible alignment input formats beyond MAPS. As a next step, we are ourselves planning to implement ScoreWarp-panels within the performance analysis tool "Listen Here!" in order to make temporal differences between performance renditions (tempo, rubato, …) assessed by the tool's users more immediately apparent.

We expect that visualizing timed information using a warped score in the way proposed here will provide for a more intuitive understanding of empirical data measured in the contexts of music listening and/or performance (including extra-musical observations, e.g., physiological measurements such as heart rate, respiratory behaviour, and galvanic skin response) relative to the prior approaches described in Section 2. However, this intuition should be tested thoroughly, both in terms of the specific use-case and of the type and complexity of music employed.

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